

Name: Queen Adecive

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TRANSPORT PHENOMENA II, CHE 312

EXAM 2

CLOSED BOOK, TIME: 75 MINUTES

Solid species A in the shape of a sphere of radius R is submerged in a large body of stagnant liquid B . Species A is slightly soluble in the liquid B with an equilibrium concentration of c_{A0} at $r = R$. The dissolved species A diffuses radially in the liquid phase while it undergoes a first-order chemical reaction.

Answer the following questions in terms of the following constants: radius of the solid sphere R , the diffusivity D_{AB} , and the rate constant k_1''' .

- Using a shell mole balance for species A , derive an ordinary differential equation for the steady-state concentration profile of species A in the liquid phase. Assume $x_A \ll 1$
- Solve the above differential equation and express the concentration profile of species A in terms of the Thiele modulus $\phi \equiv \sqrt{\frac{k_1'''}{D_{AB}}} R$ and two integration constants. Hint: $\frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = \frac{d^2}{dr^2} (rc_A)$
- Determine the integration constants using two appropriate boundary conditions for the concentration profile.
- Determine how many moles of A react per unit time.

1. $\frac{d}{dt} \frac{\partial C_A}{\partial t} = - \int_s ds \langle N_A | n \rangle + \int_v dv R_A$
Steady State

$$0 = - \int_s ds \langle N_A | n \rangle + \int_v dv R_A$$

$$0 = [N_A r 4\pi r^2]_r - [N_A r 4\pi r^2]_{r+\Delta r} + R_A 4\pi r^2 \Delta r$$

$$R_A = -k_1''' C_A$$

→ divide by volume ($4\pi r^2 \Delta r$)

$$0 = \frac{[N_A r 4\pi r^2]_r - [N_A r 4\pi r^2]_{r+\Delta r}}{4\pi r^2 \Delta r} + \frac{-k_1''' C_A 4\pi r^2 \Delta r}{4\pi r^2 \Delta r}$$

→ take limit as $\Delta r \rightarrow 0$

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$$-\frac{1}{r^2} \frac{d(N_A r^2)}{dr} - K_i C_A = 0$$

Flux

$$N_{A,r} = X_A C \left(N_{A,r} / + \left(N_{B,r} \right) \right) - D_{AB} \frac{dC_A}{dr}$$

$X_A \ll 1$ So there's no
convection term

$$N_{A,r} = -D_{AB} \frac{dC_A}{dr}$$

Plug in $N_{A,r}$

$$-\frac{1}{r^2} \frac{d}{dr} \left(-D_{AB} \frac{dC_A}{dr} r^2 \right) - K_i C_A = 0$$

Bring out constants

$$D_{AB} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dC_A}{dr} \right) - K_i C_A = 0$$

$$\frac{1}{r} D_{AB} \frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dC_A}{dr} \right) - K_i C_A = 0$$

$$\frac{1}{r} D_{AB} \frac{d^2}{dr^2} (r C_A) - K_i C_A = 0$$

Bring constants to other side

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$$2. D_{AB} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dC_A}{dr} \right) - K_1''' C_A = 0$$

$$\frac{d^2}{dr^2} (r C_A) = \frac{K_1''' C_A}{D_{AB}} r$$

$$\psi = \sqrt{\frac{K_1'''}{D_{AB}}} R$$

→ multiply both sides by $R^2 \rightarrow$ constant

$$R^2 \frac{d^2}{dr^2} (r C_A) = \frac{K_1''' R^2 C_A r}{D_{AB}}$$

$$\psi^2$$

$$\frac{d^2}{dr^2} (r C_A) = \psi^2 C_A \frac{r}{R^2}$$

move to the other side

$$\boxed{\frac{1}{R^2} \frac{d^2}{dr^2} (r^2 C_A) = \psi^2 C_A} \text{ which we can use the appendix C}$$

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$$C_A = \frac{C_{A0} R}{\sinh \phi R} \frac{1}{r} \sinh \phi r$$

$$A. \quad N_{Ar} \cdot A = -D_{AB} \frac{dC_A}{dr} \cdot 4\pi r^2 \Delta r$$

$$\frac{dC_A}{dr} = -\frac{C_1}{r^2} \cosh \phi r + \frac{C_1 \phi}{r} \sinh \phi r - \frac{C_2}{r^2} \sinh \phi r + \frac{C_2 \phi}{r} \cosh \phi r$$

Plug in B.C

$$C_1 = 0$$

$$C_2 = \frac{C_{A0} R}{\sinh \phi R}$$

$$\frac{dC_A}{dr} = -\frac{C_{A0} R}{\sinh \phi R} \frac{1}{r^2} \sinh \phi r + \frac{C_{A0} R}{\sinh \phi R} \frac{\phi}{r} \cosh \phi r$$

So

$$-D_{AB} \frac{dC_A}{dr} \cdot 4\pi r^2 \Delta r = \left(-D_{AB} \left(-\frac{C_{A0} R}{\sinh \phi R} \frac{1}{r^2} \sinh \phi r + \frac{C_{A0} R}{\sinh \phi R} \frac{\phi}{r} \cosh \phi r \right) \cdot 4\pi r^2 \Delta r \right)$$